UnionPay Integrated Circuit Card Specifications
— Product Specifications

Part VI Dual-currency Low-value Payment Specification Based on Debit/Credit Application

Version 2014

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Table of Contents

Summary of Revisions

The change listed below is associated with the current version.

Description of Change	Where to look
Added: "Card CVM limit" in Table 1	5.1
Added: "ECash Secondary Card CVM Limit" in Table 2	5.1
Added: Updated "Electronic Cash Load Threshold Value" to "Electronic Cash resets Threshold Value".	5.1
Added: Card CVM limit description	5.1

1 Application Scope

This book applies to all UPI participants.

2 Reference Documents

UnionPay Integrated Circuit Card Specifications – Basic Specification

3 Terms and Definitions

This section utilizes the following terms and definitions:

3.1 Cardholder Verification Method(CVM)

Verifies whether the cardholder a legal method and terminal, used to ensure the card was not stolen or lost.

3.2 Clearing

The handling process the issuer conducts on the transaction data submitted by the Acquirer. For the issuer, clearing is an opportunity for additional synchronization between the background counter and card counter. For the Acquirer, clearing means to send transaction data to the issuer.

3.3 Command

A message send to the IC card by the terminal. This message activates an operation or request response.

3.4 Cryptogram

The result of encryption algorithms.

3.5 Electronic Cash(E-Cash)

One kind of application with low-value payment based on debit/credit application.

3.6 Dual Currency Electronic Cash

One approach to low-value payments based on debit/credit application. Configure two types of currency code on the card, with the first currency and second currency. Conduct transaction through the terminal assigning a currency code. The transaction process is basically the same as traditional electronic cash.

3.7 Electronic Cash Balance

A counter, which expresses the balance that can be offline purchase. The approved electronic cash transaction is subtracted from the authorized total.

3.8 Electronic Cash Funds Limit

Card data that expresses the highest amount the cardholder can use in offline purchase.

3.9 Function

A handling process with one or several commands. Its operating results are used to complete all or a portion of transactions.

3.10 Issuer Action Code

Selection actions taken by the issuer based on TVR content.

3.11 Load

The process of increasing the balance of ECash in the card. Load has numerous approaches to complete the process. It is possible to transfer in the amount from a main account, or cash can be deposited, or an amount can be transferred from other accounts. The electronic cash amount after the load, however, cannot exceed the electronic cash funds limit.

3.12 Magstripe

A magnetic stripe which can store data & information.

3.13 Response

The message returned to the terminal after the IC card processing finishes and a command message is received.

3.14 Script

The command or series of commands sent to the terminal by the issuer, with the goal of inputting continual commands to the IC card.

3.15 Transaction Log

Records the latest information details, in which one can review transaction history.

3.16 Terminal

Equipment installed at the transaction point to complete the finance transaction, and used on the communication to the IC card. This includes interface equipment, and can also include other module and interfaces such as the interface with the host communication.

3.17 Unload

Unload means the process of reducing electronic cash amounts to 0. Unload results in transferring all electronic cash account amounts to the card's main account or an equivalent amount in cash to the cardholder.

4 Symbols and Abbreviations

ARPC	Authorization Response Cryptogram
ARQC	Authorization Request Cryptogram
ATC	Application Transaction Counter
ATM	Automated Teller Machine
ECash	Electronic Cash
EMV	Europay MasterCard VISA
CDOL	Card Risk Management Data Object List
CAM	Card Authentication Method
CVM	Cardholder Verification Method
IACs	Issuer Action Code
PIN	Personal Identification Number
TC	Transaction Certificate

5 Dual Currency Cash Technology

The current electronic cash card can only support one currency, one electronic cash balance, and relevant risk control parameters. The market now requires the existence of an electronic cash card product that supports dual currency. Its basic principles are based on existing UICS electronic cash transaction process and security mechanism definitions. A second currency is saved on the card with relevant electronic cash data. When conducting a transaction, corresponding selection is made based on transaction currency code data to conduct risk control and balance update. There are no special processing demands on the terminal during this process. For the issuer backend, differentiation of corresponding electronic cash accounts through transaction currency code is a requirement.

These standards only offer description of dual currency electronic cash card and single currency electronic cash card transaction differences. For other sections please refer to Part V and Part VI of the UICS Specifications.

The dual currency electronic cash in qUICS during transaction defined in these standards are only applicable to "Supporting Low-value Review" options (card additional processing options, Tag9F68). For the interim, the "Support of Low-value and CTTA Review" as well as "Support of low-value or CTTA Review" options will not be considered.

5.1 Dual Currency Electronic Cash Card New Data Elements

On the foundation of traditional electronic cash, dual currency electronic cash as established and added related data elements for the second currency electronic cash. Details are as follows:

Single currency electronic cash data elements are as found in Table 1.

Table 1 Single Currency Electronic Cash Data Elements

Name of Data Element	Tag	Length	Format
Application Currency Code	9F51	2	n4
ECash Balance	9F79	6	n12
ECash Balance Limit	9F77	6	n12
ECash Issuer Authorization Code	9F74	6	a

Name of Data Element	Tag	Length	Format
ECash Single Transaction Limit	9F78	6	n12
ECash Reset Threshold	9F6D	6	n12
Card CVM limit	9F6B	6	n12

Added data elements for dual currency electronic cash are found below:

Table 2 Second Currency Electronic Cash Data Elements

Name of Data Element	Tag	Length	Format
ECash Secondary Application Currency Code	DF71	2	n4
ECash Secondary Application Balance	DF79	6	n12
ECash Secondary Application Balance Limit	DF77	6	n12
ECash Secondary Application Single Transaction Limit	DF78	6	n12
ECash Secondary Application Reset Threshold	DF76	6	n12
ECash Secondary Card CVM Limit	DF72	6	n12

Parameters for the first and second currencies only represent different currency types. The meaning of the parameters is the same and is therefore not explained any further.

- Electronic Cash Balance: This variable saves the remaining balance that can be used when offline purchase. For each successful electronic cash transaction, a corresponding amount will be subtracted from the authorized amount. When the authorized amount exceeds the electronic cash balance, it is recommended that transaction be completed through an online authorization method.

- Electronic Cash Fund Limit: The highest total fund amount that the cardholder can use in offline purchase in electronic cash transactions applications. It is also the limit that can be reached by card loading. The issuer can adjust this limit.

- Electronic Cash Issuer Authorization Code: The code used to identify and approve electronic cash transaction codes. During offline transaction approval, this code is saved to the clearing packet in a format of "ECashCxxx". In this, xxx is the issuer defined code.

- Electronic Cash Single Transaction Limit: The single electronic cash authorized limit on the card, used to control single transaction electronic cash transaction risk. When personalized, this is entered by the issuer and is reconfigured by the issuer.

- Electronic Cash resets Threshold Value: Triggers the card to conduct automatic loading within the limits of the available balance. When the card's offline available balance is lower than this threshold value, the card will request to go online and automatically conduct loading.

- Card CVM limit: The maximum value can be used in ECash application without implementing cardholder verification. And ECash Secondary Card CVM Limit (tag 'DF72') only exists in dual currency Ecash qUICS transactions.

5.2 Dual Electronic cash card and traditional electronic cash transaction differences in standard debit applications

This section only describes differences between dual currency electronic cash and traditional electronic cash transactions in the contact interface. For areas not covered please refer to *UnionPay Integrated Circuit Card Specifications*.

5.2.1 Personalization Requests

In addition to new data element personalization, during personalize PDOL must include electronic cash terminal support indicator (Tag 9F7A), authorized amount (Tag 9F02), and transaction currency code (Tag 5F2A).

5.2.2 Initialization Application

According to the definitions in *UnionPay Integrated Circuit Card Specifications*, the card decides if a transaction is an electronic cash transaction based on a series of conditions in the Get Processing Option commands. For dual currency electronic cash card, with the exception of special handling required to decide whether or not the transaction currency code and application currency code match, other conditions are the same as in the standards.

Dual currency cash card process for determining if the transaction currency code and the application currency code match in the Get Processing Option command are as below:

After the card obtains a transaction currency code (Tag 5F2A), comparison is first conducted between the first currency electronic cash application currency code (Tag9F51). If they match, activation occurs and processing of all data elements of the first currency is conducted. If it does not match, further comparison of the second currency electronic cash application currency code is conducted. If they match, activation occurs and processing of all data elements of the second currency is conducted. If there is no match, then it will be processed based on existing processes according to the non-matching currency. Based on transaction currency code activated first currency or second currency data elements, further handling processes are kept the same as existing processes.

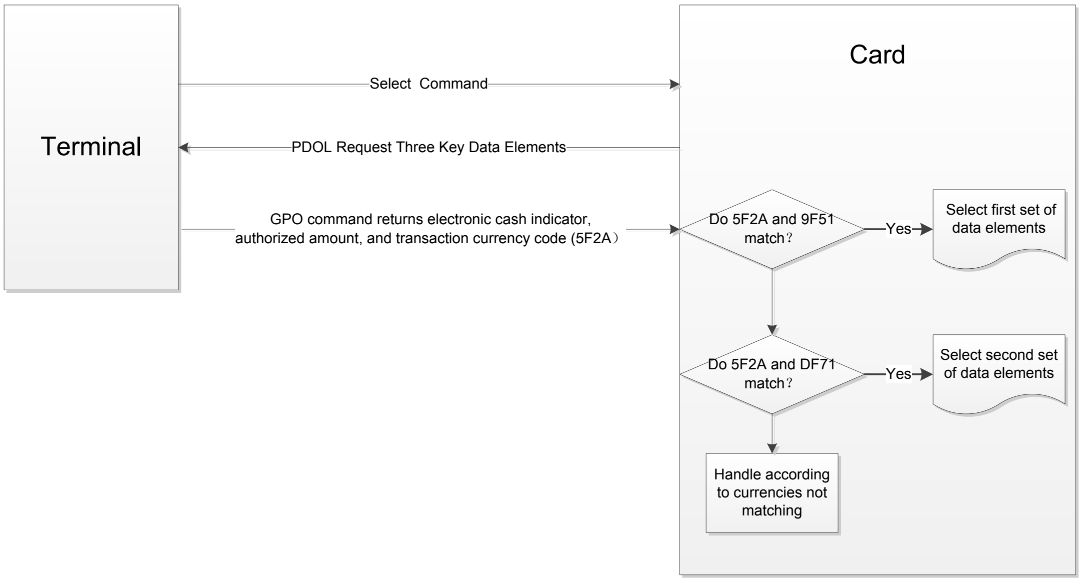

Figure 1 Dual Currency Electronic Cash Consumer Transaction Process

5.2.3 GET DATA Command

Based on the definitions in UICS, when the terminal receives AIP and AFL, and the electronic cash issue authorization code (Tag 9F74) exists, then electronic cash balance (9F79) and electronic cash reconfiguration threshold value will be read through GET DATA command, to be used in the processing of terminal action analysis.

For dual currency electronic cash cards, when Get Processing Option receives identical transaction currency code the same as second currency code (DF71), then:

When the terminal utilizes GET DATA command to read electronic cash balance (9F79), the card should return the DF79 value;

When the terminal utilizes GET DATA command to read electronic cash reconfiguration threshold (9F6D), the card should return the DF76 value

When the terminal utilizes GET DATA command to read electronic cash balance limit (9F77), the card should return the DF77 value;

When the terminal utilizes GET DATA command to read electronic cash single transaction limit (9F78), the card should return the DF78 value;

When the terminal utilizes GET DATA command to read offline available amount (9F5D), the card should utilize the DF79 value in calculation.

For dual currency electronic cash cards, when GPO received transaction currency code and second currency could application currency code (DF71) are different or no GPO command is received, then the card will handle processing according to the first currency when handling GET DATA commands. The objective of this design is to ensure that assuming terminal processes and commands have not changed, the card is able to correctly identify whether the terminal using GET DTA command wishes to read the first set of data elements or the second set of data elements.

5.2.4 Card Action Analysis

Second currency card action analysis is the same as the process for the first currency. The area with difference is as follows: for dual currency electronic cash cards, CDOL1 and CDOL2 need to include transaction currency code (5F2A). When the Generate AC command receives a different transaction currency code from Get Processing Option command, then the card should refuse the transaction.

When the card is responding to Generate AC commands, the response data includes issue application data, and 9F10 might include electronic cash balance. For dual currency cash cards, whether to response with the first currency electronic cash balance (9F79) or second electronic cash balance (DF79) should be decided according to transaction currency code. In addition, when the transaction approved, update the corresponding currency electronic cash balance.

5.2.5 Issuer Script Processing

The card must support issuer editing of second currency electronic cash data through issuer script command PUT DATA. After the card receives PUT DATA commands, editing of corresponding data is conducted according to Tag, and has no special handling. For example if the issuer needs to edit the first currency electronic cash balance, then PUT DATA command P1 P2 parameters will be 9F79, and the card will directly edit 9F79 corresponding values. If the issuer needs to edit second currency electronic cash balance, PUT DATA command P1 P2 parameters will be DF79, and the card directly edits DF79 corresponding values.

5.3 Dual Currency Cash Card Differences between traditional electronic cash transactions in qUICS

This section only describes dual currency electronic cash card qUICS transaction process differences as compared with traditional electronic cash transactions. For similarities, please refer to *UnionPay Integrated Circuit Card Specifications*

5.3.1 Personalized Requests

The card needs to request transaction currency code (Tag 5F2A) at PDOL, and therefore it is not suitable to utilize cipher text version 17 online qUICS. This is because in this case PDOL does not request transaction currency code. If PDOL does not include transaction currency code 5F2A, then handling will be completed according to exiting processes.

Other personalized requirements: Card additional processing options (9F68) Byte 1 bit 7 “supports micro and CTTA review” and Byte1 bit6 “supports micro or CTTA review” must be configured to 0. If 9F68 has configured non-low-value review, then the second currency transaction will be handled as if the first currency does not match.

The issuer application data (9F10) issuer customized data cannot include the following:

Cumulative Total Transaction Amount (CTTA) IDD ID of 0x02;

Electronic cash balance and CTTA, IDD ID of 0x03;

CTTA and CTTAL, IDD ID of 0x04.

The above personalized requests need to be rigorously reviewed when the issuer supplies the card to the bank card testing center for personalized data verification.

5.3.2 Card Processing Upon Receipt of GPO Commands

After the card obtains transaction currency code (Tag 5F2A), first it will conduct comparison with the first currency electronic cash application currency code (Tag 9F51). If they match, activation occurs and it conducts processing of the first currency matching set of data elements. If they don’t match, conduct comparison with the second currency electronic cash application currency code. If they match, activate and conduct processing of the second currency matching set of data elements. If they don't match, then processing will be handled according to the existing processes upon not matching.

The process for currency matching and data element selection should occur prior to card risk management, or at the first stage of risk management "configuration of currency matching or not matching". After activating the first transaction currency code or the second currency data elements, following handling processes and exiting processes remain the same. See Figure 2

Two data that require special attention:

9F5D: When the card needs to return 9F5D (offline available balance), the card should automatically select the use of 9F79 or DF79 according to currency match results then use it in calculating with 9F5D.

9F10: When the card needs to return electronic cash balance (9F79) issuer customized data, or offline available balance (9F5D), the card should automatically select the use of 9F79 or DF79 according to currency match results.

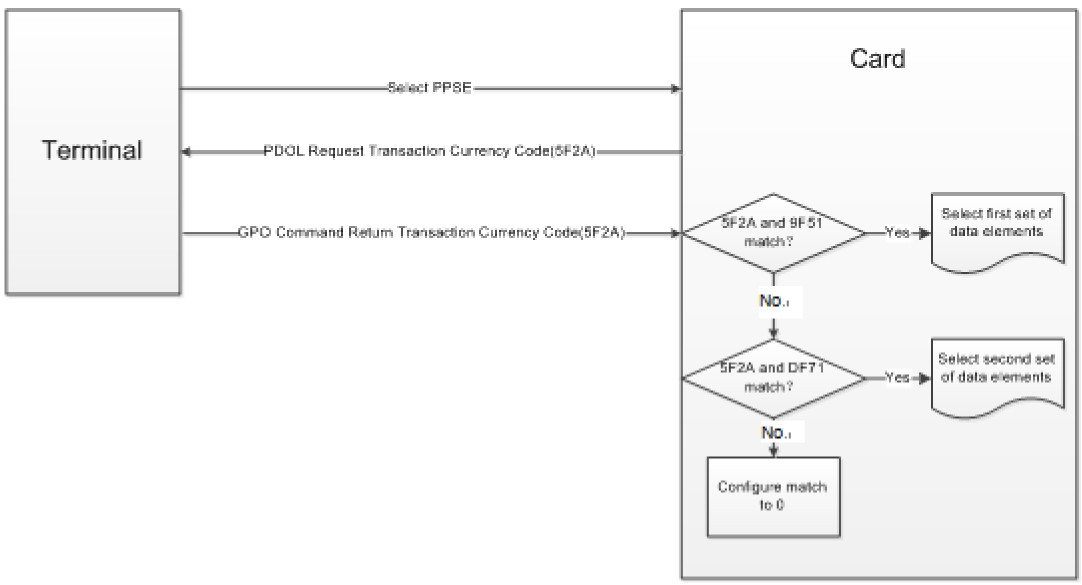

Figure 2 qUICS Dual Currency Cash Consumer Transaction Process

5.3.3 GET DATA Command

For dual currency electronic cash card, when GPO received transaction currency code and second currency code (DF71) are the same, then:

When the terminal utilizes GET DATA command to read electronic cash balance (9F79), the card should return the DF79 value;

When the terminal utilizes GET DATA command to read electronic cash reconfiguration threshold (9F6D), the card should return the DF76 value

When the terminal utilizes GET DATA command to read electronic cash balance limit (9F77), the card should return the DF77 value;

When the terminal utilizes GET DATA command to read electronic cash single transaction limit (9F78), the card should return the DF78 value;

When the terminal utilizes GET DATA command to read offline available amount (9F5D), the card should utilize the DF79 value in calculation.

For dual currency electronic cash cards, when GPO received transaction currency code and second currency could application currency code (DF71) are different or no GPO command is received, then the card will handle processing according to the first currency when handling GET DATA commands. The objective of this design is to ensure that assuming terminal processes and commands have not changed, the card is able to correctly identify whether the terminal using GET DTA command wishes to read the first set of data elements or the second set of data elements.

5.4 Card Commands

This section describes the following common commands that are different for dual currency cash card as opposed to traditional electronic cash cards:

- Electronic cash balance inquiry command
- Update electronic cash parameter command

5.4.1 Electronic Cash Balance Inquiry

Electronic cash balance inquiry command allows the terminal to directly read card offline consumer balance.

Use GET DATA command APDU to read electronic cash balance. This command and corresponding data formats are as found in the table below:

Table 3 Electronic Cash Balance Inquiry GET DATA Command

Bytes	Value	Notes
CLA	‘80’	
INS	‘CA’	

Bytes	Value	Notes
P1		'9F79'is the first currency electronic cash balance data unit Tag 'DF79' is the second currency electronic cash balance data unit Tag
P2		
LC	N/A	
Data	N/A	
Le	'00'	

Table 4 Electronic Cash Balance Inquiry Response

Bytes	Value	Notes
Tag(T)	'9F79 / DF79'	
Length(L)	'06'	6 bytes
Data(V)	Electronic cash balance	Expressed by application definition currency
SW1/SW2		Status information

Electronic cash balance is expressed by application definition currency.

5.4.2 Update Electronic Cash Parameter Command

The electronic cash parameter on the card can be updated through UICS standard "PUT DATA" command. The issuer mainframe when authorizing responses uses script to create and send this command. It is possible to utilize script to command the updated card electronic cash parameter first currency and second currency electronic cash balance; electronic cash balance limit, electronic cash single transaction limit, and electronic cash reconfiguration threshold.

Table 5 Update Electronic Cash Parameter PUT DATA Command

Bytes	Value	Notes
CLA	'04'	
INS	'DA'	
P1		P1, P2 are electronic cash parameter Tag
P2		
LC	0A	
Data	value	Electronic cash parameter new value
Le	'00'	

Table 6 Electronic Cash Parameter P1, P2 values

Data element name	P1	P2
First currency electronic cash balance	9F	79
First currency electronic cash balance limit	9F	77
First currency electronic cash single transaction limit	9F	78
First currency electronic cash reconfiguration threshold	9F	6D
Second currency electronic cash balance	DF	79
Second currency electronic cash balance limit	DF	77
Second currency electronic cash single transaction limit	DF	78

Data element name	P1	P2
Second currency electronic cash reconfiguration threshold	DF	76

Successful update of variable response status word is SW1SW2='9000'. For other possible error status definitions please refer to UICS debit application card standards.